SDN: Protocol Identifier

Mininet:

```
vboxuser@UbuntuVM:~$ sudo python3 topology.py
*** Creating network
*** Adding controller
Unable to contact the remote controller at 127.0.0.1:6653
*** Adding hosts:
h1 h2 h3 h4 h5
*** Adding switches:
s1
*** Adding links:
(10.00Mbit 5ms delay) (10.00Mbit 5ms delay) (h1, s1) (10.00Mbit 5ms delay) (10.00Mbit 5ms delay) (h2, s1) (10.00Mbit 5ms delay) (10.00Mbit 5ms delay) (
3, s1) (10.00Mbit 5ms delay) (10.00Mbit 5ms delay) (h4, s1) (10.00Mbit 5ms delay) (10.00Mbit 5ms delay) (h5, s1)
*** Configuring hosts
h1 h2 h3 h4 h5
*** Starting controller
c0
*** Starting 1 switches
s1 ...(10.00Mbit 5ms delay) (10.00Mbit 5ms delay) (10.00Mbit 5ms delay) (10.00Mbit 5ms delay) (10.00Mbit 5ms delay)

=============== TOPOLOGY RUNNING ===============
Switch: s1
Hosts: h1, h2, h3, h4, h5
IP Range: 10.0.0.1 - 10.0.0.5
================================================

Try these commands in CLI:
pingall
h1 iperf -s &
h2 iperf -c 10.0.0.1
h3 ping -c 3 h4
*** Starting CLI:
mininet>
```

OS-kin:

```
vboxuser@UbuntuVM:~$ source osken-env/bin/activate
(osken-env) vboxuser@UbuntuVM:~$ osken-manager controller.py
2 RLock(s) were not greened, to fix this error make sure you run eventlet.monkey_patch() before importing any other modules.
loading app controller.py
loading app os_ken.controller.ofp_handler
instantiating app controller.py of TrafficClassifier
instantiating app os_ken.controller.ofp_handler of OFPHandler
[CONNECTED] Switch 1
```

ICMP:

```
mininet> h1 nping --icmp -c 20 10.0.0.2

Starting Nping 0.7.94SVN ( https://nmap.org/nping ) at 2026-04-21 12:01 UTC
SENT (0.0681s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=1] IP [ttl=64 id=53089 iplen=28 ]
RCVD (0.2266s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=1] IP [ttl=64 id=23220 iplen=28 ]
SENT (1.0682s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=3] IP [ttl=64 id=53089 iplen=28 ]
RCVD (1.1197s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=3] IP [ttl=64 id=23811 iplen=28 ]
SENT (2.0690s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=3] IP [ttl=64 id=53089 iplen=28 ]
RCVD (2.0935s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=3] IP [ttl=64 id=24299 iplen=28 ]
SENT (3.0716s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=4] IP [ttl=64 id=53089 iplen=28 ]
RCVD (3.1124s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=4] IP [ttl=64 id=24440 iplen=28 ]
SENT (4.0727s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=5] IP [ttl=64 id=53089 iplen=28 ]
RCVD (4.1010s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=5] IP [ttl=64 id=24592 iplen=28 ]
SENT (5.0871s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=6] IP [ttl=64 id=53089 iplen=28 ]
RCVD (5.1869s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=6] IP [ttl=64 id=25103 iplen=28 ]
SENT (6.0934s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=7] IP [ttl=64 id=53089 iplen=28 ]
RCVD (6.1377s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=7] IP [ttl=64 id=25127 iplen=28 ]
SENT (7.1030s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=8] IP [ttl=64 id=53089 iplen=28 ]
RCVD (7.2058s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=8] IP [ttl=64 id=25152 iplen=28 ]
SENT (8.1107s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=9] IP [ttl=64 id=53089 iplen=28 ]
RCVD (8.1415s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=9] IP [ttl=64 id=25778 iplen=28 ]
SENT (9.1144s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=10] IP [ttl=64 id=53089 iplen=28 ]
RCVD (9.1441s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=10] IP [ttl=64 id=26776 iplen=28 ]
SENT (10.1297s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=11] IP [ttl=64 id=53089 iplen=28 ]
RCVD (10.2054s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=11] IP [ttl=64 id=27206 iplen=28 ]
SENT (11.1378s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=12] IP [ttl=64 id=53089 iplen=28 ]
RCVD (11.1665s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=12] IP [ttl=64 id=27600 iplen=28 ]
SENT (12.1437s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=13] IP [ttl=64 id=53089 iplen=28 ]
RCVD (12.1726s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=13] IP [ttl=64 id=27624 iplen=28 ]
SENT (13.1452s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=14] IP [ttl=64 id=53089 iplen=28 ]
RCVD (13.2157s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=14] IP [ttl=64 id=27945 iplen=28 ]
SENT (14.1534s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=15] IP [ttl=64 id=53089 iplen=28 ]
RCVD (14.1949s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=15] IP [ttl=64 id=28613 iplen=28 ]
SENT (15.1553s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=16] IP [ttl=64 id=53089 iplen=28 ]
RCVD (15.1987s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=16] IP [ttl=64 id=29404 iplen=28 ]
SENT (16.2223s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=17] IP [ttl=64 id=53089 iplen=28 ]
RCVD (16.2484s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=17] IP [ttl=64 id=30140 iplen=28 ]
SENT (17.2280s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=18] IP [ttl=64 id=53089 iplen=28 ]
RCVD (17.2533s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=18] IP [ttl=64 id=30582 iplen=28 ]
SENT (18.2355s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=19] IP [ttl=64 id=53089 iplen=28 ]
RCVD (18.2634s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=19] IP [ttl=64 id=30819 iplen=28 ]
SENT (19.2425s) ICMP [10.0.0.1 > 10.0.0.2 Echo request (type=8/code=0) id=39942 seq=20] IP [ttl=64 id=53089 iplen=28 ]
RCVD (19.2698s) ICMP [10.0.0.2 > 10.0.0.1 Echo reply (type=0/code=0) id=39942 seq=20] IP [ttl=64 id=31446 iplen=28 ]

Max rtt: 158.448ms | Min rtt: 24.503ms | Avg rtt: 50.246ms
Raw packets sent: 20 (560B) | Rcvd: 20 (560B) | Lost: 0 (0.00%)
Nping done: 1 IP address pinged in 19.30 seconds
```

```
[CONNECTED] Switch 1
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] ICMP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] ICMP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[SUMMARY] ICMP → Total Packets: 40
```

TCP:

```
mininet> h1 nping --tcp -c 20 10.0.0.2

Starting Nping 0.7.94SVN ( https://nmap.org/nping ) at 2026-04-21 12:03 UTC
SENT (0.0285s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (0.0858s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0
SENT (1.0435s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (1.0772s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0
SENT (2.0448s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (2.0709s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0
SENT (3.0455s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (3.1599s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0
SENT (4.0471s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (4.0731s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0
SENT (5.0560s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (5.0880s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0
SENT (6.0640s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (6.0899s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0
SENT (7.0670s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (7.0909s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0
SENT (8.0698s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (8.0944s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0
SENT (9.0782s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (9.1014s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0
SENT (10.0792s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (10.1023s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0
SENT (11.0803s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (11.1066s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0
SENT (12.0823s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (12.1087s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0
SENT (13.0896s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (13.1214s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0
SENT (14.0909s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (14.1183s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0
SENT (15.0966s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (15.1303s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0
SENT (16.0996s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (16.1283s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0
SENT (17.1014s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (17.1369s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0
SENT (18.1053s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (18.1365s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0
SENT (19.1061s) TCP 10.0.0.1:7752 > 10.0.0.2:80 S ttl=64 id=47564 iplen=40  seq=2888844412 win=1480
RCVD (19.1354s) TCP 10.0.0.2:80 > 10.0.0.1:7752 RA ttl=64 id=0 iplen=40  seq=0 win=0

Max rtt: 114.387ms | Min rtt: 23.084ms | Avg rtt: 33.918ms
Raw packets sent: 20 (800B) | Rcvd: 20 (800B) | Lost: 0 (0.00%)
Nping done: 1 IP address pinged in 19.18 seconds
```

```
[PACKET] TCP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] TCP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] TCP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] TCP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] TCP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] TCP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] TCP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] TCP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] TCP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] TCP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] TCP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] TCP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] TCP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] TCP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] TCP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] TCP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] TCP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] TCP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] TCP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] TCP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] TCP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] TCP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] TCP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] TCP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] TCP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] TCP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] TCP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] TCP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] TCP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] TCP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] TCP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] TCP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] TCP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] TCP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] TCP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] TCP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[PACKET] TCP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] TCP | 00:00:00:00:00:02 -> 00:00:00:00:00:01
[SUMMARY] TCP → Total Packets: 40
```

UDP:

```
mininet> h2 iperf -s -u -p 5001 &
mininet> h1 nping --udp -c 20 -p 5001 10.0.0.2

Starting Nping 0.7.94SVN ( https://nmap.org/nping ) at 2026-04-21 12:05 UTC
SENT (0.0423s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28
SENT (1.0499s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28
SENT (2.0523s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28
SENT (3.0726s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28
SENT (4.0730s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28
SENT (5.0734s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28
SENT (6.0742s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28
SENT (7.0776s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28
SENT (8.0824s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28
SENT (9.0832s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28
SENT (10.0846s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28
SENT (11.0861s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28
SENT (12.0941s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28
SENT (13.0954s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28
SENT (14.0961s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28
SENT (15.1018s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28
SENT (16.1025s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28
SENT (17.1043s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28
SENT (18.1059s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28
SENT (19.1683s) UDP 10.0.0.1:53 > 10.0.0.2:5001 ttl=64 id=22353 iplen=28

Max rtt: N/A | Min rtt: N/A | Avg rtt: N/A
Raw packets sent: 20 (560B) | Rcvd: 0 (0B) | Lost: 20 (100.00%)
Nping done: 1 IP address pinged in 20.35 seconds
mininet>
```

```
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[PACKET] UDP | 00:00:00:00:00:01 -> 00:00:00:00:00:02
[SUMMARY] UDP → Total Packets: 20
```